SHARIF UNIVERSITY OF TECHNOLOGY • FALL 2026

Security and Privacy in Machine Learning

Course Project Guidelines

Course website

Overview. The course project gives you the opportunity to conduct a small research study. The teaching staff will suggest several topics and introductory papers, but students are also free to choose their own topics, as long as they relate to the course material. Whatever topic you choose, you will review previous work, develop a research question or extension, and investigate it through a study with a clear rationale. You are not expected to propose a new algorithm or outperform existing methods.

1. Phase I: Literature Review and Research Proposal (40 points)

Start by reading the main papers on your topic. If you choose a topic suggested by the teaching staff, include the recommended papers, then find additional relevant studies on your own. Your review should explain what researchers have established so far, how the main approaches compare, and what questions or limitations remain. Based on this review, identify a focused research question or extension that you can realistically investigate, and describe how you plan to study it.

Computational resources. The course cannot provide computing resources. Choose a project that you can complete using freely available resources, such as the free tiers of Google Colab or Kaggle. Keep their compute, memory, and usage limits in mind when selecting models, datasets, and experiments. Do not assume that you will have uninterrupted GPU access. Briefly explain in your proposal why the planned experiments are feasible with these resources.

Required deliverables

- **Short presentation (10–15 minutes):** introduce the topic, summarize the main approaches and findings, discuss their strengths and limitations, and present your proposed idea.
- **Reading list:** provide full citations for the main papers on your topic, including relevant papers you found independently.
- **Research proposal (1–2 pages):** explain the problem and its importance, state your research question or proposed extension, show how it builds on previous work, and describe how you plan to evaluate it.

After Phase I, the teaching staff will provide feedback on the scope and feasibility of each proposal. You may revise your plan based on this feedback, but discuss any major changes with the staff before moving forward.

2. Phase II: Experimental Study and Workshop Paper (60 points)

In Phase II, carry out your study. Implement or adapt the methods you need, design and run experiments, make appropriate comparisons, and explain what the results show. You may extend an existing method, investigate an empirical question, or reproduce earlier work and add a new experiment or analysis. Reproducing published results alone is not sufficient.

Required deliverables

- **Workshop-style paper (4–6 pages of main text):** describe your research question, related work, methods, experimental setup, results, interpretation, and limitations. References and appendices are excluded from the main-text page limit.
- **Code and reproduction instructions:** provide your code, configurations, dependencies, and clear instructions for rerunning the experiments, where feasible.

Clearly distinguish your own contributions from work done by others. Include limitations and unsuccessful experiments in your report rather than omitting them.

3. Grading

Criterion	Points
Phase I: Literature Review and Research Proposal	**40**
Breadth and quality of the literature review	15
Comparison of prior work and identification of open questions	10
Research question and motivation	10
Presentation clarity and organization	5
Phase II: Experimental Study and Workshop Paper	**60**
Correctness of implementation and validity of experiments	20
Experimental design and choice of baselines	15
Analysis and interpretation of results	15
Paper quality and reproducibility of code and results	10
Base total	**100**

4. Research Outcomes and Bonus Credit

Negative results are welcome when the evidence supports them. Your grade will reflect the reasoning behind your study, the quality of your experiments, and how clearly you explain your findings. You do not need to improve on a baseline to earn full credit. A carefully investigated idea that does not work can still receive full marks. Incomplete or unreliable experiments will be graded based on the work demonstrated.

You can earn up to **15 bonus points** for work beyond the basic requirements: up to 5 points for an original and important research question, up to 5 for a useful finding backed by strong evidence, and up to 5 for particularly insightful analysis or a substantial extension. Negative findings are also eligible. The maximum project score is therefore **115/100**.

5. General Expectations

- Treat the suggested papers as a starting point, not a complete reading list. You are responsible for selecting your research methods and explaining your choices.
- Support your claims with evidence and use baseline comparisons or controls where helpful. Be clear about the limits of your conclusions. Choosing larger models or more complex methods does not, by itself, lead to a higher grade.
- Follow the course's academic integrity and ethical security-research policies. Do not conduct unauthorized attacks or experiments against real systems, services, or individuals.

Topic selection, team arrangements, deadlines, submission instructions, and paper template will be announced separately.